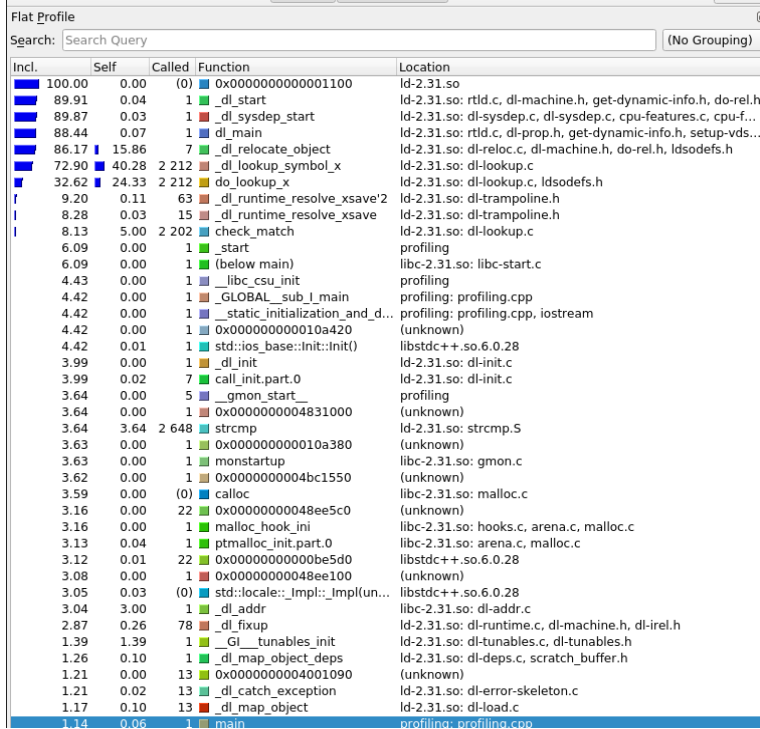


IVS - profiling

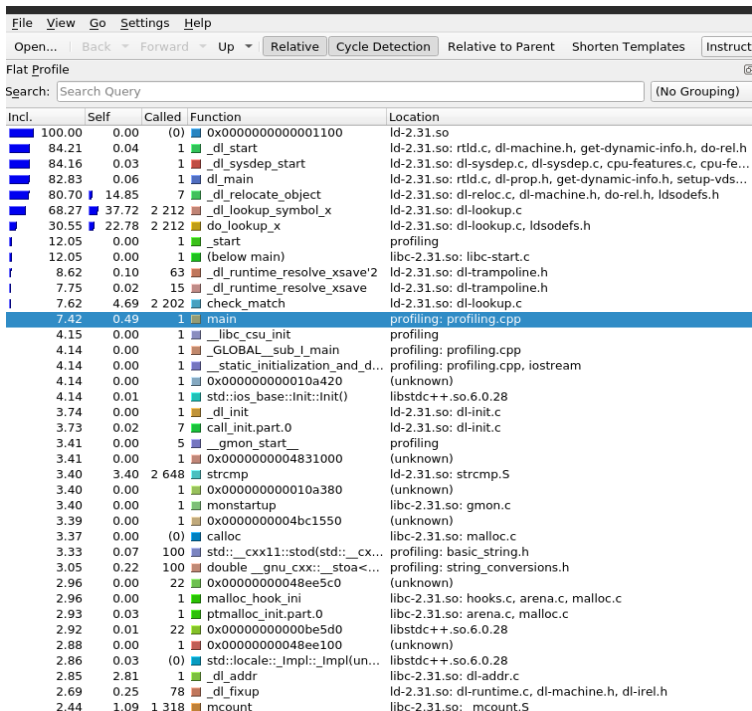
Used profiler: callgrind

In this picture is the output of callgrind (shown with kcachegrind) with input consisting of 10 numbers.



Incl.	Self	Called	Function	Location
100.00	0.00	(0)	0x0000000000001100	ld-2.31.so
89.91	0.04	1	_dl_start	ld-2.31.so: rtld.c, dl-machine.h, get-dynamic-info.h, do-rel.h
89.87	0.03	1	_dl_sysdep_start	ld-2.31.so: dl-sysdep.c, dl-sysdep.c, cpu-features.c, cpu-f...
88.44	0.07	1	dl_main	ld-2.31.so: rtld.c, dl-prop.h, get-dynamic-info.h, setup-vds...
86.17	15.86	7	_dl_relocate_object	ld-2.31.so: dl-reloc.c, dl-machine.h, do-rel.h, ldsdefs.h
72.90	40.28	2 212	_dl_lookup_symbol_x	ld-2.31.so: dl-lookup.c
32.62	24.33	2 212	do_lookup_x	ld-2.31.so: dl-lookup.c, ldsdefs.h
9.20	0.11	63	_dl_runtime_resolve_xsave'2	ld-2.31.so: dl-trampoline.h
8.28	0.03	15	_dl_runtime_resolve_xsave	ld-2.31.so: dl-trampoline.h
8.13	5.00	2 202	check_match	ld-2.31.so: dl-lookup.c
6.09	0.00	1	_start	profiling
6.09	0.00	1	(below main)	libc-2.31.so: libc-start.c
4.43	0.00	1	_libc_csu_init	profiling
4.42	0.00	1	_GLOBAL__sub_I_main	profiling: profiling.cpp
4.42	0.00	1	_static_initialization_and_d...	profiling: profiling.cpp, iostream
4.42	0.00	1	0x000000000010a420	(unknown)
4.42	0.01	1	std::ios_base::Init::Init()	libstdc++.so.6.0.28
3.99	0.00	1	_dl_init	ld-2.31.so: dl-init.c
3.99	0.02	7	call_init.part.0	ld-2.31.so: dl-init.c
3.64	0.00	5	_gmon_start__	profiling
3.64	0.00	1	0x00000000004831000	(unknown)
3.64	3.64	2 648	strcmp	ld-2.31.so: strcmp.S
3.63	0.00	1	0x000000000010a380	(unknown)
3.63	0.00	1	monstartup	libc-2.31.so: gmon.c
3.62	0.00	1	0x00000000004bc1550	(unknown)
3.59	0.00	(0)	calloc	libc-2.31.so: malloc.c
3.16	0.00	22	0x000000000048ee5c0	(unknown)
3.16	0.00	1	malloc_hook_ini	libc-2.31.so: hooks.c, arena.c, malloc.c
3.13	0.04	1	ptmalloc_init.part.0	libc-2.31.so: arena.c, malloc.c
3.12	0.01	22	0x00000000000be5d0	libstdc++.so.6.0.28
3.08	0.00	1	0x000000000048ee100	(unknown)
3.05	0.03	(0)	std::locale::_Impl::_Impl(un...	libstdc++.so.6.0.28
3.04	3.00	1	_dl_addr	libc-2.31.so: dl-addr.c
2.87	0.26	78	_dl_fixup	ld-2.31.so: dl-runtime.c, dl-machine.h, dl-irel.h
1.39	1.39	1	_GI__tunables_init	ld-2.31.so: dl-tunables.c, dl-tunables.h
1.26	0.10	1	_dl_map_object_deps	ld-2.31.so: dl-deps.c, scratch_buffer.h
1.21	0.00	13	0x00000000004001090	(unknown)
1.21	0.02	13	_dl_catch_exception	ld-2.31.so: dl-error-skeleton.c
1.17	0.10	13	_dl_map_object	ld-2.31.so: dl-load.c
1.14	0.06	1	main	profiling: profiling.cpp

In this one the input consists of 100 numbers.



Incl.	Self	Called	Function	Location
100.00	0.00	(0)	0x0000000000001100	ld-2.31.so
84.21	0.04	1	_dl_start	ld-2.31.so: rtld.c, dl-machine.h, get-dynamic-info.h, do-rel.h
84.16	0.03	1	_dl_sysdep_start	ld-2.31.so: dl-sysdep.c, dl-sysdep.c, cpu-features.c, cpu-f...
82.83	0.06	1	dl_main	ld-2.31.so: rtld.c, dl-prop.h, get-dynamic-info.h, setup-vds...
80.70	14.85	7	_dl_relocate_object	ld-2.31.so: dl-reloc.c, dl-machine.h, do-rel.h, ldsdefs.h
68.27	37.72	2 212	_dl_lookup_symbol_x	ld-2.31.so: dl-lookup.c
30.55	22.78	2 212	do_lookup_x	ld-2.31.so: dl-lookup.c, ldsdefs.h
12.05	0.00	1	_start	profiling
12.05	0.00	1	(below main)	libc-2.31.so: libc-start.c
8.62	0.10	63	_dl_runtime_resolve_xsave'2	ld-2.31.so: dl-trampoline.h
7.75	0.02	15	_dl_runtime_resolve_xsave	ld-2.31.so: dl-trampoline.h
7.62	4.69	2 202	check_match	ld-2.31.so: dl-lookup.c
7.42	0.49	1	main	profiling: profiling.cpp
4.15	0.00	1	_libc_csu_init	profiling
4.14	0.00	1	_GLOBAL__sub_I_main	profiling: profiling.cpp
4.14	0.00	1	_static_initialization_and_d...	profiling: profiling.cpp, iostream
4.14	0.00	1	0x000000000010a420	(unknown)
4.14	0.01	1	std::ios_base::Init::Init()	libstdc++.so.6.0.28
3.74	0.00	1	_dl_init	ld-2.31.so: dl-init.c
3.73	0.02	7	call_init.part.0	ld-2.31.so: dl-init.c
3.41	0.00	5	_gmon_start__	profiling
3.41	0.00	1	0x00000000004831000	(unknown)
3.40	3.40	2 648	strcmp	ld-2.31.so: strcmp.S
3.40	0.00	1	0x000000000010a380	(unknown)
3.40	0.00	1	monstartup	libc-2.31.so: gmon.c
3.39	0.00	1	0x00000000004bc1550	(unknown)
3.37	0.00	(0)	calloc	libc-2.31.so: malloc.c
3.33	0.07	100	std::cxx11::stod(std::cx...	profiling: basic_string.h
3.05	0.22	100	double__gnu_cxx::__stoa<...	profiling: string_conversions.h
2.96	0.00	22	0x000000000048ee5c0	(unknown)
2.96	0.00	1	malloc_hook_ini	libc-2.31.so: hooks.c, arena.c, malloc.c
2.93	0.03	1	ptmalloc_init.part.0	libc-2.31.so: arena.c, malloc.c
2.92	0.01	22	0x00000000000be5d0	libstdc++.so.6.0.28
2.88	0.00	1	0x000000000048ee100	(unknown)
2.86	0.03	(0)	std::locale::_Impl::_Impl(un...	libstdc++.so.6.0.28
2.85	2.81	1	_dl_addr	libc-2.31.so: dl-addr.c
2.69	0.25	78	_dl_fixup	ld-2.31.so: dl-runtime.c, dl-machine.h, dl-irel.h
2.44	1.09	1 318	mcount	libc-2.31.so: mcount.S

And here of 1000 numbers

Incl.	Self	Called	Function	Location
100.00	0.00	(0)	0x0000000000001100	ld-2.31.so
51.41	0.03	1	_dl_start	ld-2.31.so: rtld.c, dl-machine.h, get-dynamic-info.h, do-rel.h
51.39	0.02	1	_dl_sysdep_start	ld-2.31.so: dl-sysdep.c, dl-sysdep.c, cpu-features.c, cpu-fe...
50.56	0.04	1	dl_main	ld-2.31.so: rtld.c, dl-prop.h, get-dynamic-info.h, setup-vds...
49.26	9.07	7	_dl_relocate_object	ld-2.31.so: dl-reloc.c, dl-machine.h, do-rel.h, ldsdefs.h
46.30	0.00	1	_start	profiling
46.30	0.00	1	(below main)	libc-2.31.so: libc-start.c
43.48	2.99	1	main	profiling: profiling.cpp
41.67	23.02	2 212	_dl_lookup_symbol_x	ld-2.31.so: dl-lookup.c
20.47	0.46	1 007	std::_cxx11::stod(std::_cx...	profiling: basic_string.h
18.76	1.35	1 007	double __gnu_cxx::_stoa<...	profiling: string_conversions.h
18.65	13.91	2 212	do_lookup_x	ld-2.31.so: dl-lookup.c, ldsdefs.h
14.78	6.60	13 109	mcount	libc-2.31.so: _mcount.S
11.73	2.13	1 008	nth_power(double, double)	profiling: calc_funcs.cpp
11.15	0.12	1 007	strtod	libc-2.31.so: strtod.c
11.03	5.90	1 007	__strtod_internal	libc-2.31.so: strtod_l.c
8.18	8.18	13 109	_mcount_internal	libc-2.31.so: mcount.c
6.09	1.55	4 033	compare_double(double, d...	profiling: calc_funcs.cpp
5.26	0.06	63	_dl_runtime_resolve_xsave'2	ld-2.31.so: dl-trampoline.h
4.73	0.01	15	_dl_runtime_resolve_xsave	ld-2.31.so: dl-trampoline.h
4.65	2.86	2 202	check_match	ld-2.31.so: dl-lookup.c
3.46	0.36	1 009	is_int(double)	profiling: calc_funcs.cpp
2.80	0.53	2 014	plus(double, double)	profiling: calc_funcs.cpp
2.53	0.00	1	_libc_csu_init	profiling
2.53	0.00	1	_GLOBAL__sub_I_main	profiling: profiling.cpp
2.53	0.00	1	_static_initialization_and_d...	profiling: profiling.cpp, iostream
2.52	0.00	1	0x0000000000010a420	(unknown)
2.52	0.01	1	std::ios_base::Init::Init()	libstdc++.so.6.0.28
2.49	0.05	1 007	0x0000000000010a3e0	(unknown)
2.49	0.23	4 897	0x0000000000010a3f0	(unknown)

Summary:

Our functions are starting to show only with bigger inputs. As we can see in the 3rd screenshot, the 'worst' function is `nth_power`, which isn't surprising, since it is our longest function and it's rather ineffective. The problem lies within our implementation of over/underflow checking.